

Olive oil intake and CHD in the European Prospective Investigation into Cancer and Nutrition Spanish cohort.

Olive oil is well known for its cardioprotective properties; however, epidemiological data showing that olive oil consumption reduces incident CHD events are still limited. Therefore, we studied the association between olive oil and CHD in the European Prospective Investigation into Cancer and Nutrition (EPIC) Spanish cohort study. The analysis included 40 142 participants (38 % male), free of CHD events at baseline, recruited from five EPIC-Spain centres from 1992 to 1996 and followed up until 2004. Baseline dietary and lifestyle information was collected using interview-administered questionnaires. Cox proportional regression models were used to assess the relationship between validated incident CHD events and olive oil intake (energy-adjusted quartiles and each 10 g/d per 8368 kJ (2000 kcal) increment), while adjusting for potential confounders. During a 10·4-year follow-up, 587 (79 % male) CHD events were recorded. Olive oil intake was negatively associated with CHD risk after excluding dietary mis-reporters (hazard ratio (HR) 0·93; 95 % CI 0·87, 1·00 for each 10 g/d per 8368 kJ (2000 kcal) and HR 0·78; 95 % CI 0·59, 1·03 for upper v. lower quartile). The inverse association between olive oil intake (per 10 g/d per 8368 kJ (2000 kcal)) and CHD was more pronounced in never smokers (11 % reduced CHD risk ($P = 0·048$)), in never/low alcohol drinkers (25 % reduced CHD risk ($P < 0·001$)) and in virgin olive oil consumers (14 % reduced CHD risk ($P = 0·072$)). In conclusion, olive oil consumption was related to a reduced risk of incident CHD events. This emphasises the need to conserve the traditional culinary use of olive oil within the Mediterranean diet to reduce the CHD burden.